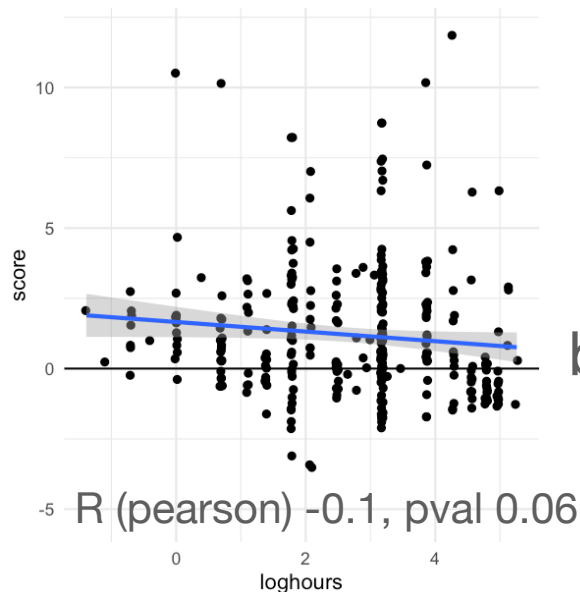
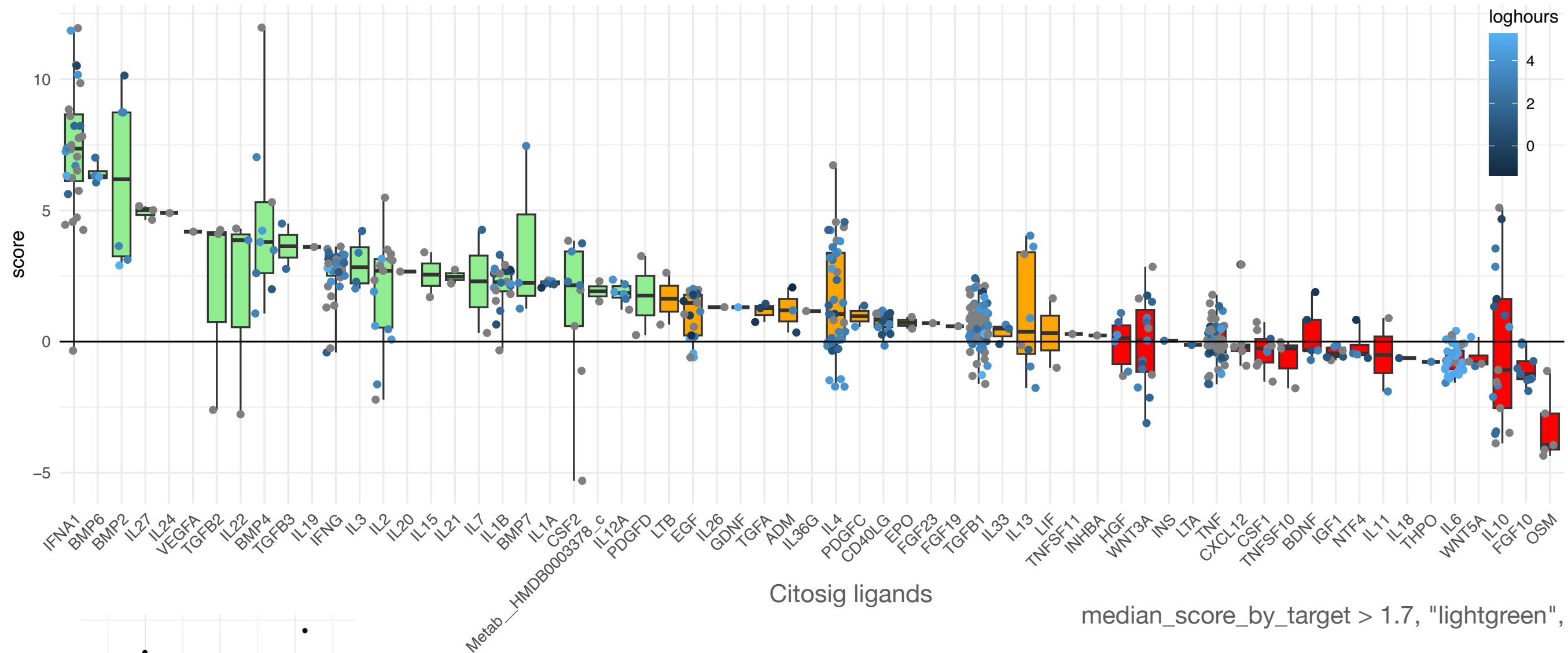


Ligand MOON scores calculated from citosig experiment

z-scores

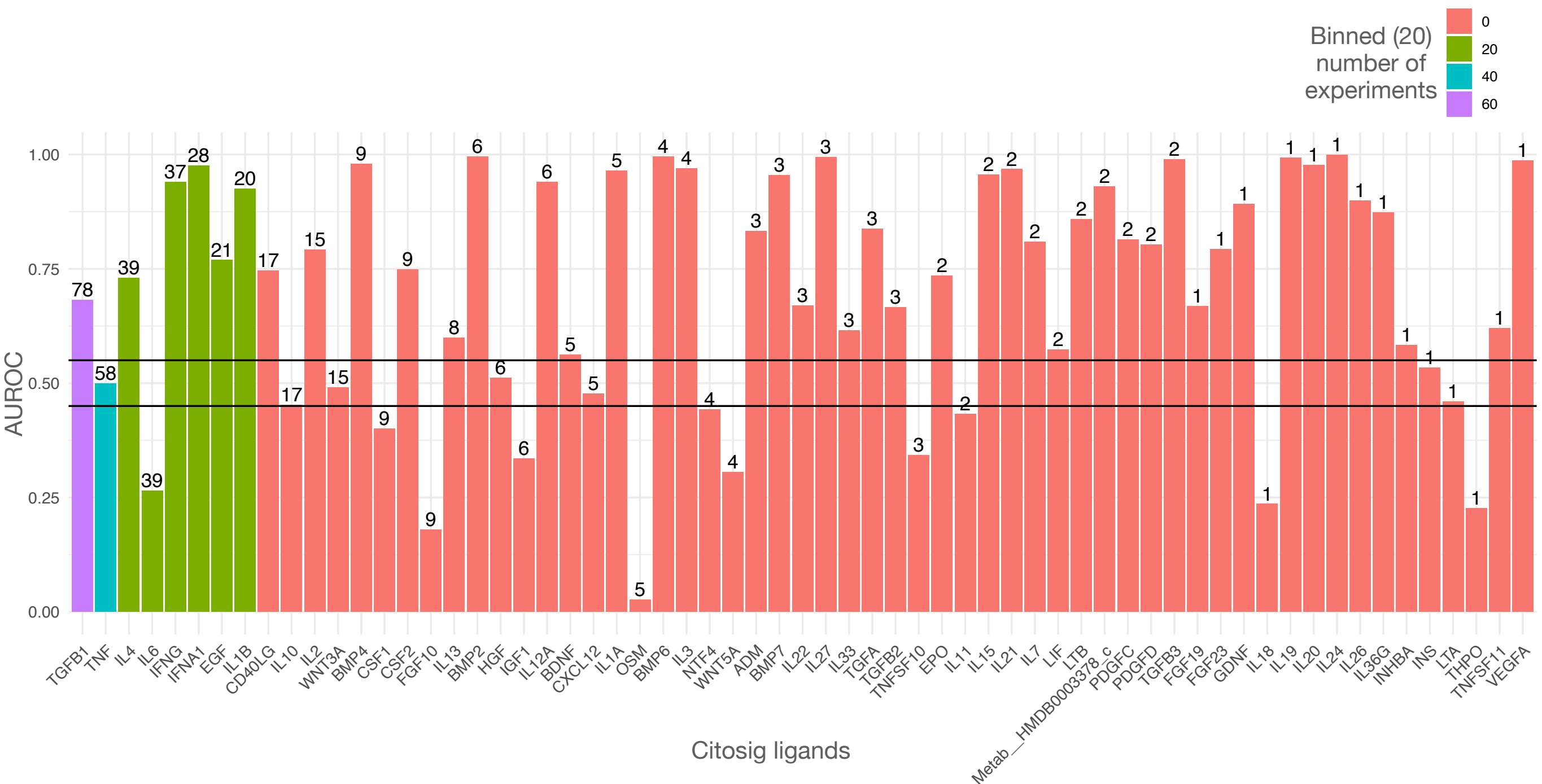


Small anti-correlation
between score and time of
collection (log hours)

median_score_by_target > 1.7, "lightgreen",
median_score_by_target > 0.3, "orange",
else, "red"

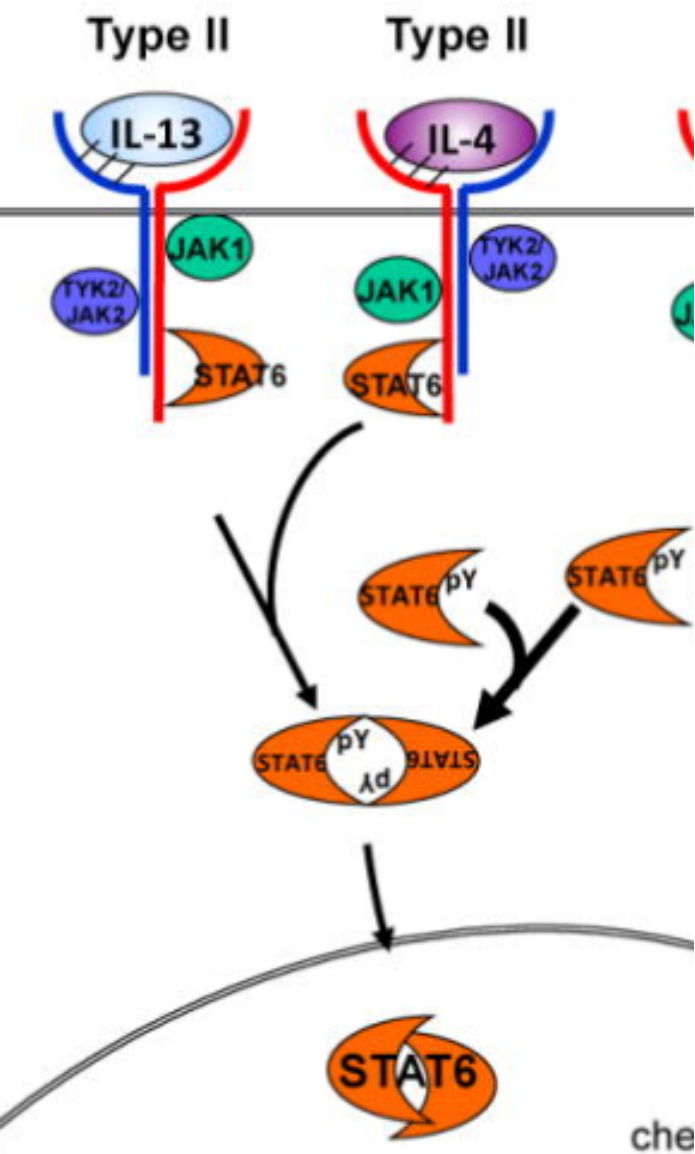
Thanks Robin for the citosig data

AUROC of ligand scores against citosig “ground truth”



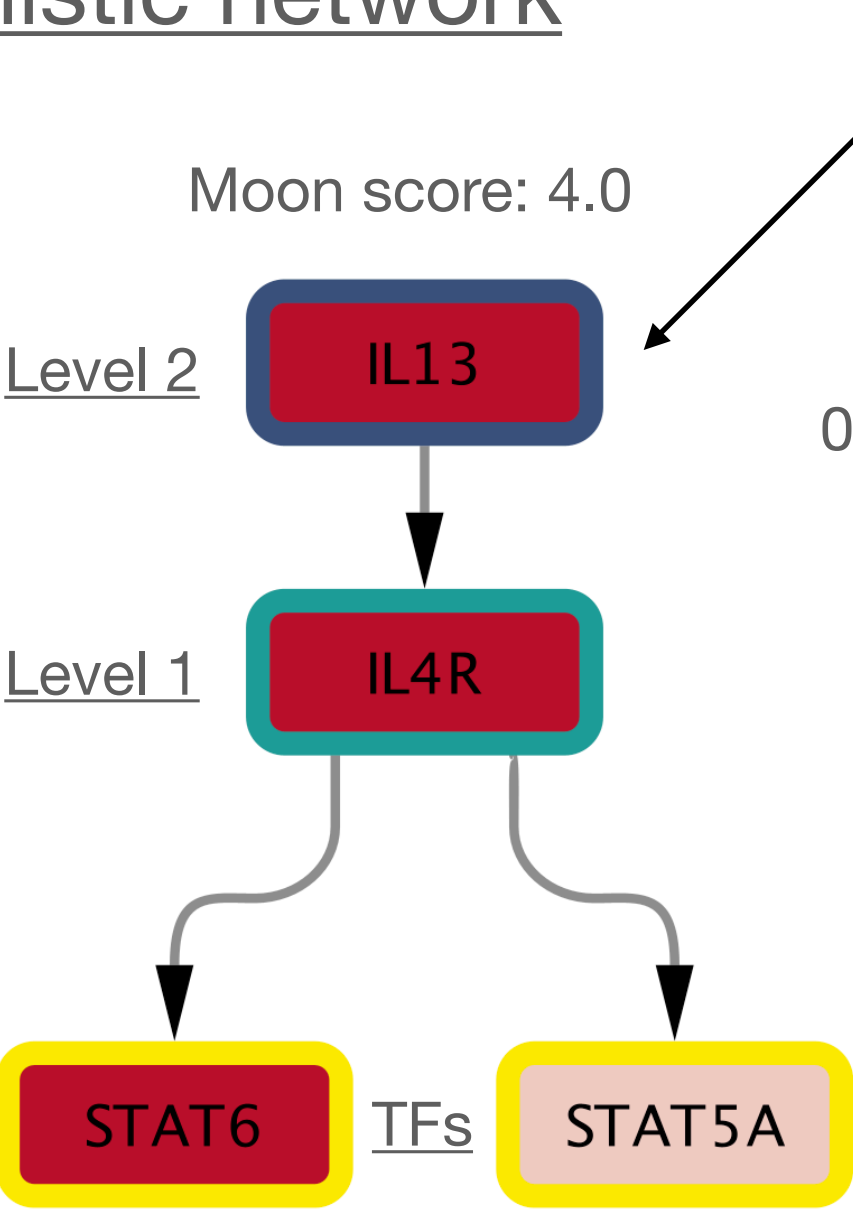
Most AUROCs are computed from a small number of experiments. As expected from the moon scores, the performances are quite heterogeneous. It requires further investigation of how we got those scores in the first place, given the data and the prior knowledge used.

Prediction of IL13 activity from downstream TFs and relevant mechanistic network

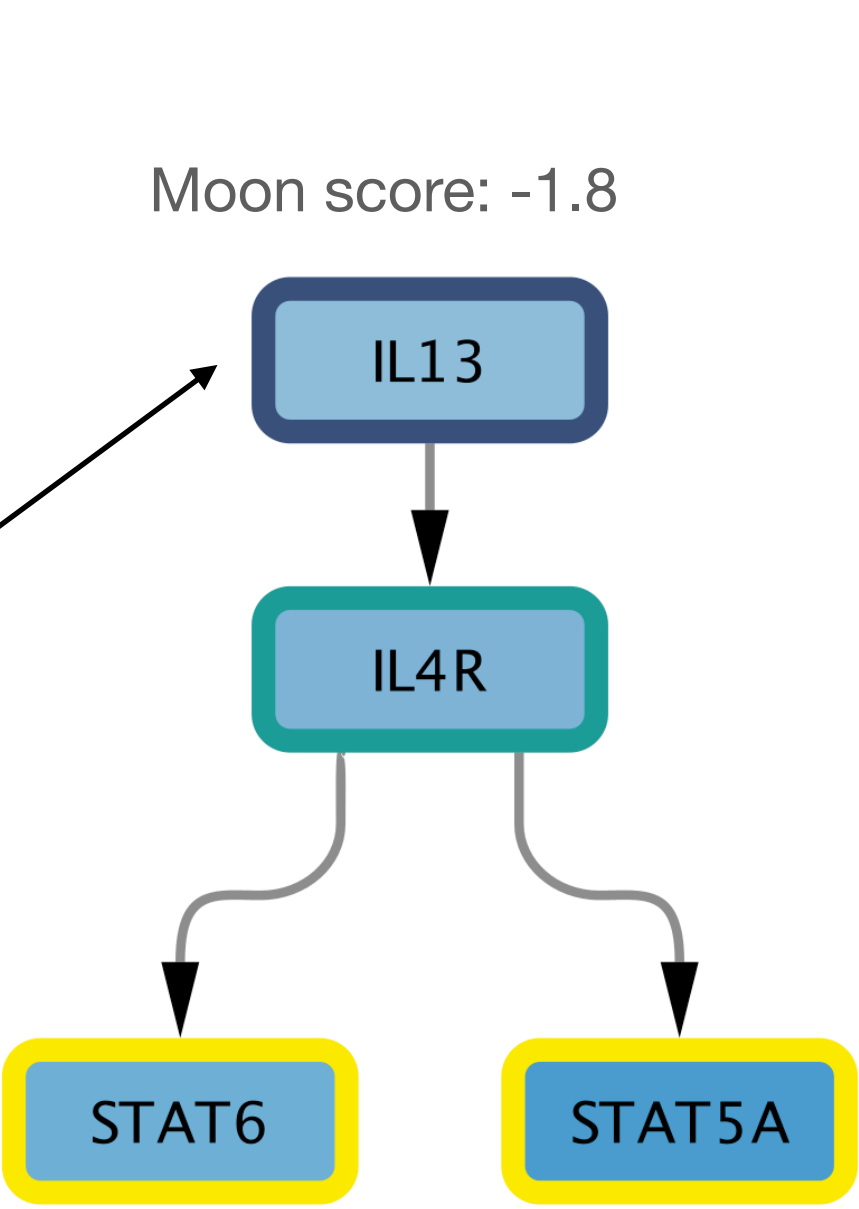


<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4546937/>

N = 8
IFNA1 AUROC: 0.59



IL13@Condition:Blood (PBMC)
&Duration:24h@GSE79027.RN
ASeq.SRP071332_GRCh38

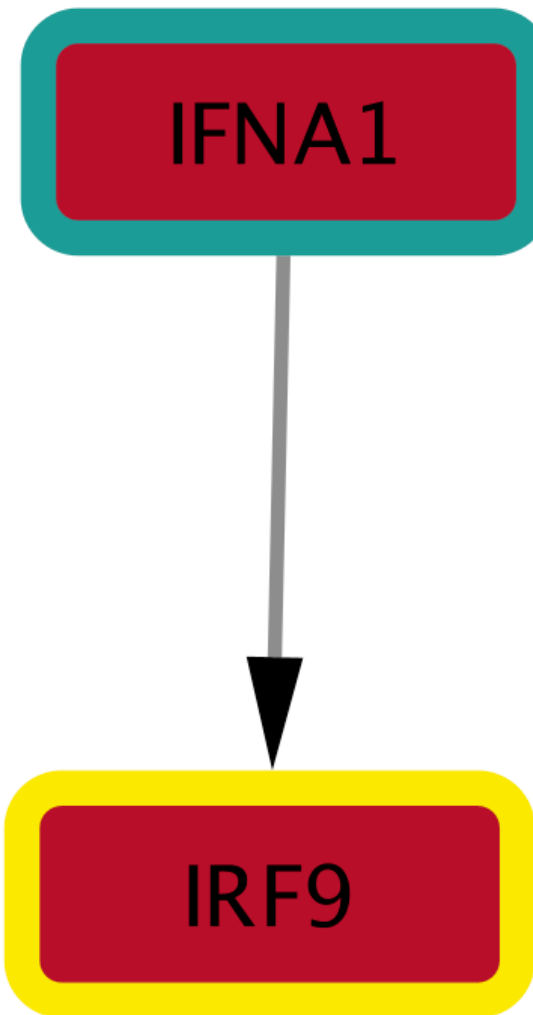


IL13@Condition:Human
bronchial cell line
A549&Duration:24h@GSE3183.
MicroArray.HG-U133A

Hard to say if poor performance is due to mistake in prior knowledge or data.

Prediction of IFNA1 activity from downstream TFs and relevant mechanistic network

N = 28
IFNA1 AUROC: 0.98

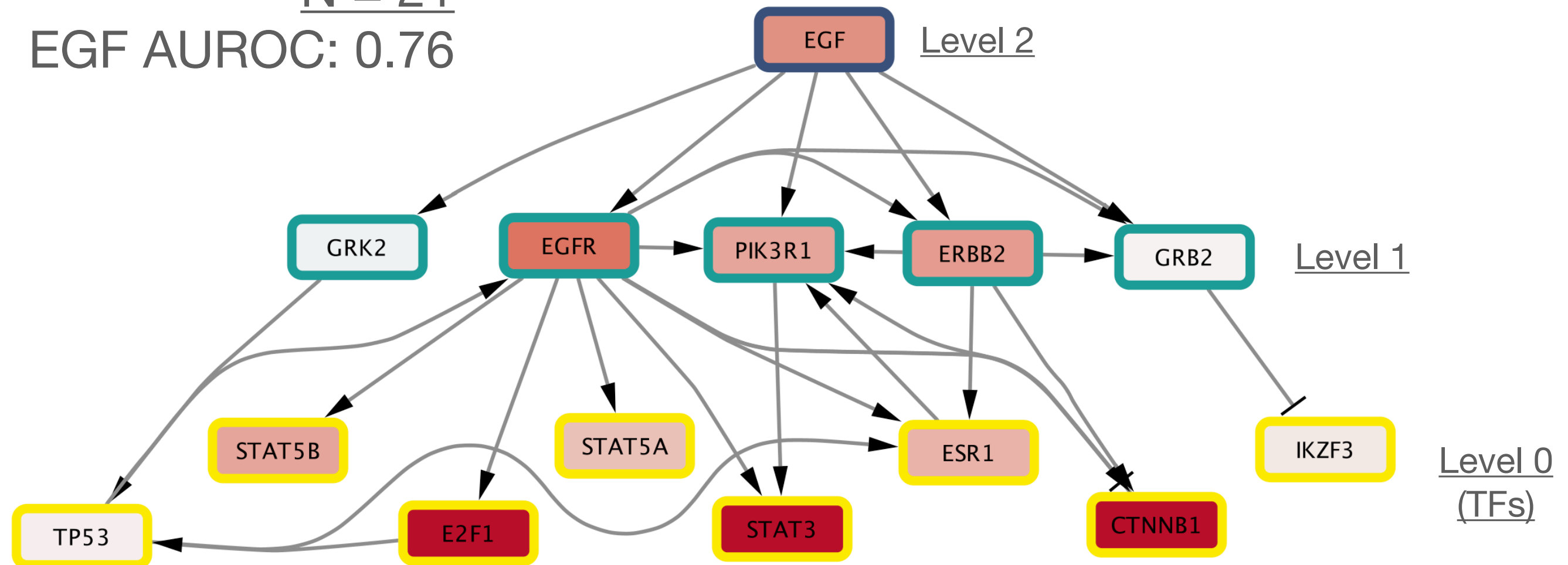


You can predict IFNA1 from IRF9 activity alone.
98% of the time, it works every time. Doesn't
seem to need a complicated model for some
ligands.

Prediction of EGF activity from downstream TFs and relevant mechanistic network

N = 21

EGF AUROC: 0.76



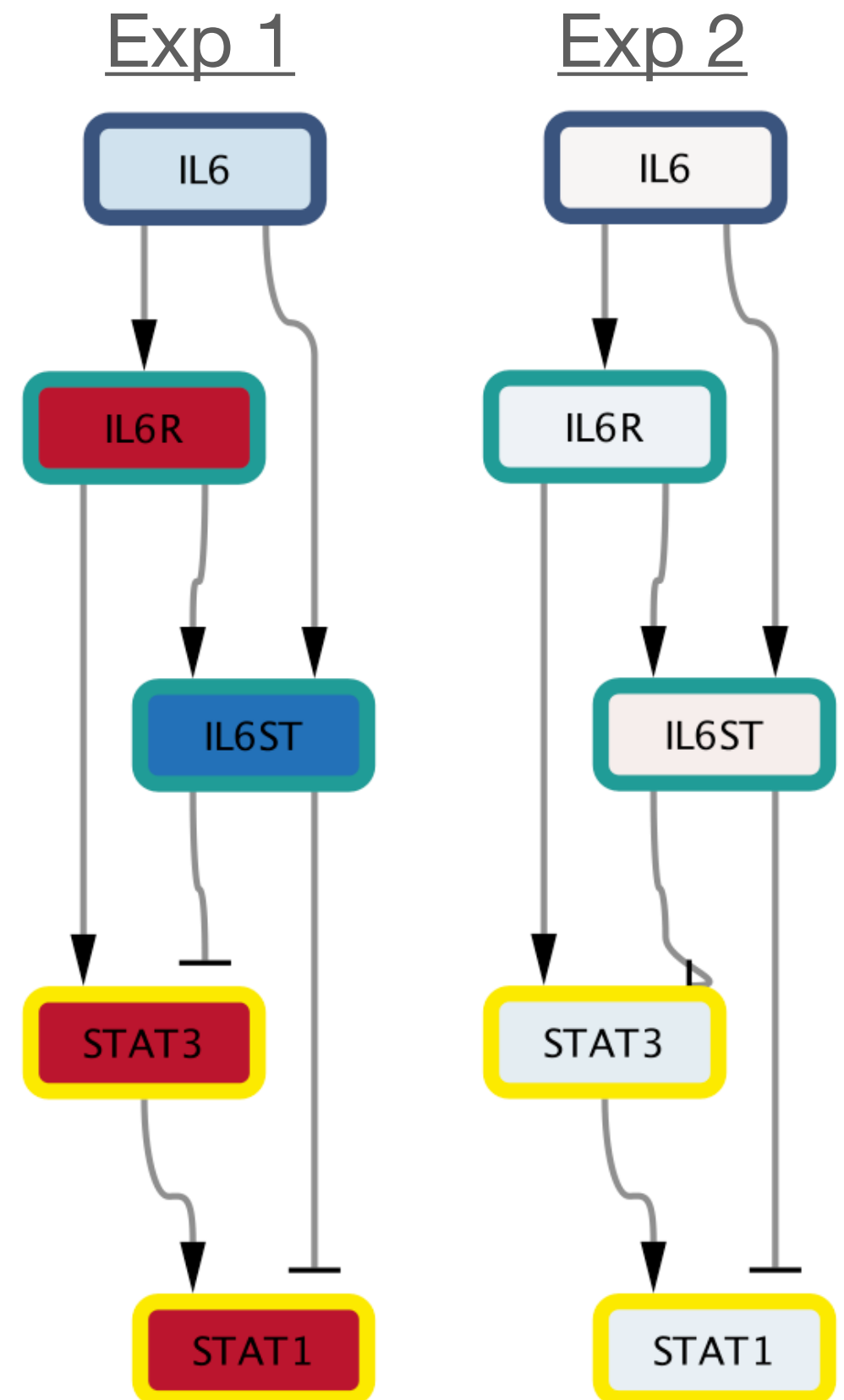
EGF is relatively well predicted from this set of TF activities. Are the connections between EGF and GRK2/GRB2 actually relevant? Very context specific? They seem to impair the prediction of EGF activity in this specific set of experiment (N=21).

Prediction of IL6 activity from downstream TFs and relevant mechanistic network

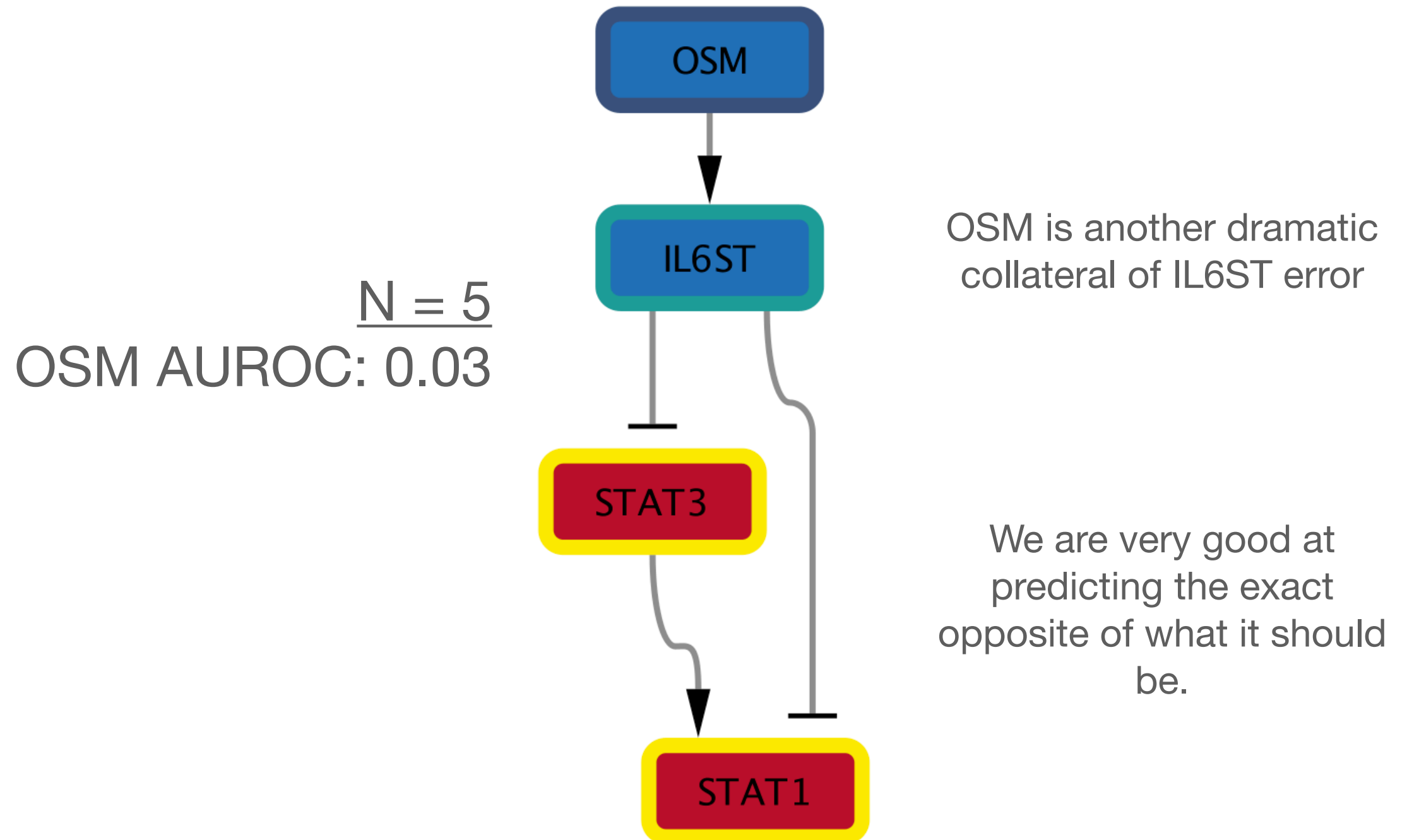
N = 39
IL6 AUROC: 0.26
IL6R AUROC: 0.55
IL6ST AUROC: 0.17

IL6ST is annotated in omnipath as an inhibitor of STAT3 and STAT1. From checking the literature, this is likely a mistake, it should be activations. Then IL6 would actually be very well predicted.

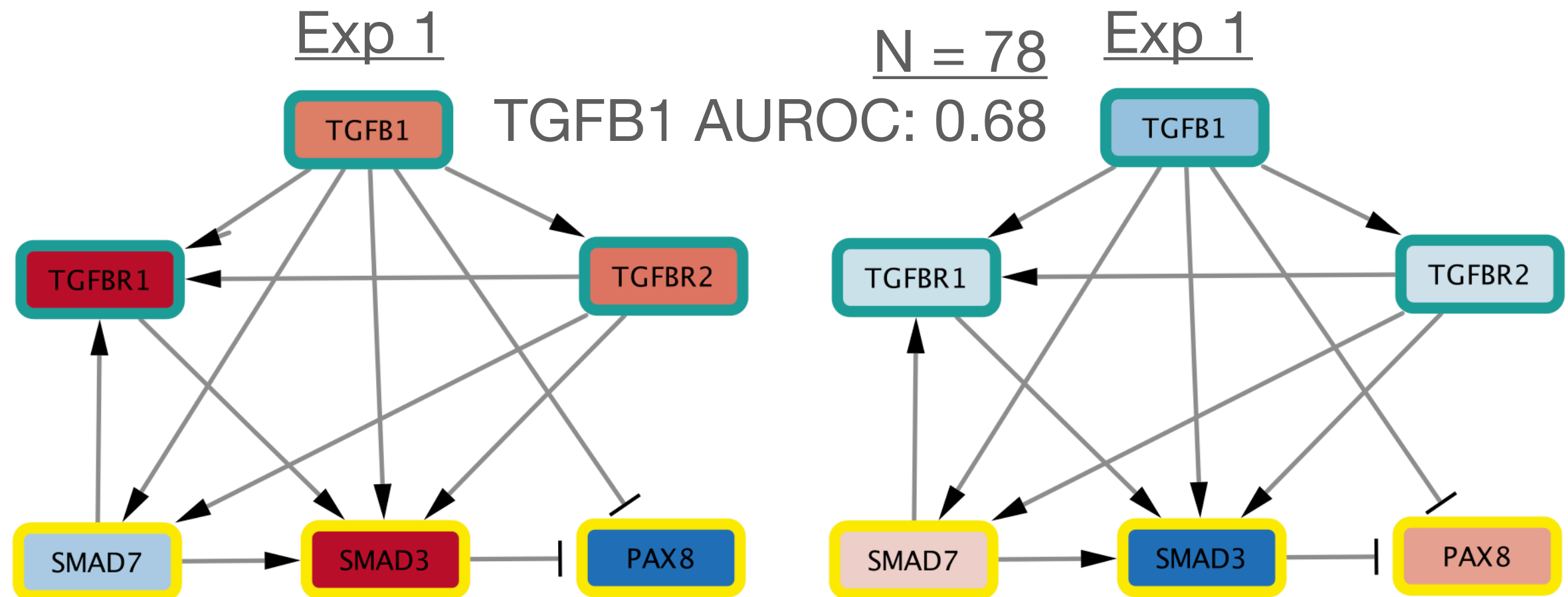
Still, in some experiments (ex right network), STAT3/1 are not activated after IL6 application, which seems to inconsistent with known biological mechanisms.



Prediction of OSM activity from downstream TFs and relevant mechanistic network



Prediction of TGFB1 activity from downstream TFs and relevant mechanistic network



TGFB1 is moderately well predicted. The performance seems to be impaired by the activation of SMAD7 by TGFB1. This is likely a wrong annotation. We can probably expect a sharp increase of performance by removing it.